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ORIGINAL ARTICLE

Tube task hand preference in captive hylobatids

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Abstract The link between laterality in humans and other primates is still hotly debated. Hylobatids have been rather neglected in this research area, yet they can provide important insights because: (1) they share with humans a complex vocal repertoire, which in humans is thought to be associated with brain hemispheric specialization and lateralized behaviors; (2) their adaptation to arboreality has produced unique postural constraints; (3) the little that is known about laterality in gibbons is contradictory (captive studies have provided conflicting results, while a field study on siamangs reported a population-level left-hand preference). To clarify this, we investigated hand preference in captive hylobatids [$n = 42$; 22 siamangs (*Symphalangus syndactylus*) and 20 gibbons (*Hylobates* sp., *Nomascus leucogenys*)] in nine Japanese facilities. We had a large sample size, controlled for possible confounds (posture, enclosure limitations) and used a well-established testing protocol (tube task). Handedness indices calculated from raw frequencies and bouts were highly correlated and showed a significant left-hand skew, which is consistent with data from wild siamangs. Major differences between captive and wild siamangs were a larger number of ambiguously handed individuals, and no significant age-related variation in captivity. The use of the index finger elicited a much more strongly lateralized response than the thumb. These results confirmed a left-hand preference in siamangs, but were equivocal in other hylobatids, and suggest selective pressures that may have acted on the highly arboreal hylobatids to favor handedness. Our study also indicates factors that might explain the discrepancy in the literature between handedness studies on captive and wild primate populations.

Keywords Laterality · Siamang · Postural origin hypothesis · Arboreality · Thumb

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Introduction

Despite the recent increase in studies of hand preferences in non-human primates, the relationship between laterality in humans and other species remains a topic of intense debate, with many competing hypotheses addressing both proximate and ultimate causation (Frost 1980; MacNeilage et al. 1987; Vallortigara and Rogers 2005; Hopkins and Cantalupo 2008; MacNeilage et al. 2009; Rogers et al. 2013). Some authors suggest that pronounced lateralization in humans (i.e., 90% right-handedness) marks a discontinuity with other primates and is associated with various human-specific capacities such as language, bipedalism, tool manufacture and use (Bradshaw and Rogers 1993; Cashmore et al. 2008; Cashmore 2009; Crow 2010). Others have proposed broader theoretical views of the evolution of population-level handedness, taking into account variation across the whole phylum of vertebrates, including primates

(MacNeilage et al. 1987, 2009; Rogers and Andrew 2002). For instance, the postural origin (PO) hypothesis (MacNeilage et al. 1987) suggests that inherent positional behaviors within the order Primates influence the expression of right- and left-handedness. The PO theory specifically hypothesizes that more habitually arboreal species should exhibit increased left-handedness for manual actions (e.g., grasping a fruit) because the right is dominant for postural control. In contrast, in species that are more habitually terrestrial, there is a relaxation in the need for postural control, and an increase in right-handedness for more finely tuned tasks.

When considering nonhuman primates, three factors complicate the testing of the PO and other hypotheses. First, studies on primates offer conflicting evidence: many report hand preference at the individual level, whereas evidence for population-level results is less clear-cut (Ward and Hopkins 1993; Hook-Costigan and Rogers 1996; McGrew and Marchant 1997; Papademetriou et al. 2005; Hopkins 2006, 2013a; Hopkins et al. 2011, 2012; Meguerditchian et al. 2013). Some of the discordant results come from the comparison of captive and wild populations of the same species, and might reflect the influence of the experimental setting (e.g., McGrew and Marchant 1997; Hopkins et al. 2007).

The second limitation is that the types of tasks used to assess laterality vary considerably across studies. Several studies have shown that between- and within-species variation in handedness varies with the motor and visual spatial demands of the task (Steenhuis and Bryden 1989; Fagot et al. 1991; Fagot and Vauclair 1991; Vauclair et al. 2005; Blois-Heulin et al. 2006; Schweitzer et al. 2007; Laurence et al. 2011). For instance, tasks that measure hand preference for coordinated bimanual actions compared to unimanual tasks elicit stronger and more consistent hand preference at the individual and population levels in great apes (Fagot and Vauclair 1988; Sugiyama et al. 1993; Humle and Matsuzawa 2009; Llorente et al. 2009; Sousa et al. 2009; Chapelain et al. 2011; Hopkins et al. 2011; Lambert 2012; Llorente et al. 2011), Old World monkeys (Vauclair et al. 2005;; Schweitzer et al. 2007; Bennett et al. 2008; Zhao et al. 2010, 2012) and New World monkeys (Spinozzi et al. 1998; Lilak and Phillips 2008; Meunier and Vauclair 2007; Meguerditchian et al. 2012; Nelson et al. 2015).

A third limitation is that there are still major gaps in the literature, some of which are especially important, regarding species whose particular ecological/postural features would make them ideal subjects to explore the constraints and selective pressures acting on lateralization.

One such taxon is the small apes (Hylobatids). Hylobatids comprise several species of gibbons and the larger-sized siamang, all of which are highly arboreal primates living in Southeast Asia. These apes spend most of their lives in the upper forest canopy, seldom coming to the ground. This extreme adaptation to arboreality generates unique postural constraints. Additionally, because of their adaptations to an arboreal lifestyle, the morphology of the hylobatid hand is quite distinct among primates. The thumb is quite short relative to the length of the remaining digits, resulting in a very low opposability index, according to Napier and Tuttle (1993). Finally, given the association between brain hemispheric lateralization and complex communication [e.g., human language (Bradshaw and Rogers 1993)], it is relevant to study hylobatids, which are at the crossroads between monkeys and great apes, and share with humans a complex vocal repertoire (Chatterjee 2006). Unfortunately, few studies have been conducted on laterality in gibbons, and the few studies on captive gibbons in the literature provide mixed evidence (Table 1). In the sole study on hand preferences in wild hylobatids, siamangs (*Symphalangus syndactylus*) displayed a significant preference for the left hand in a complex manual task (collecting water from tree holes), both at individual and population levels (Morino 2011).

The goal of this study was to test the PO hypothesis by assessing hand preferences in a large sample of captive hylobatids, using a well-established, standardized method

Table 1 Previous laterality studies on hylobatids

Species	Setting	Task	*n*	Results	Reference
Symphalangus syndactylus	Captive	Food reaching	13	No preference	Heestand 1987
Hylobates lar	Captive	Mesh retrieval	6	Left-hand preference	Olson et al. 1990
S. syndactylus, H. lar, Hylobates concolor	Captive	Food reaching	19	No preference. All five adult females right-handed. Adults more lateralized than immatures	Stafford et al. 1990
S. syndactylus	Captive	Leading limb	25	No preference	Redmond and Lamperez 2004
S. syndactylus	Wild	Water collecting	49	Left-hand preference. Adults more lateralized than immatures	Morino 2011

allowing for cross-species comparison: the tube task. This task, requiring the simultaneous use of two hands, has been shown to elicit consistent hand preference at the individual and species levels (Hopkins 2013a, b). To date, the tube task has been used in 13 different primate species spanning eight genera (Hopkins 2013a), but it has never been administered to any hylobatid species. Based on the PO predictions regarding such highly arboreal primates, as well as previous results on other laterality measures (Olson et al. 1990; Morino 2011), we predicted that hylobatids would exhibit a population-level left-hand preference.

Methods

Study subjects

We tested 42 hylobatids, from seven species, in nine captive locations (Table 2; Table 5, Appendix). Age-class was determined following Palombit (1992). More information on the facilities and the history of each study subject can be found at http://www.shigen.nig.ac.jp/gain. One of the subjects (Raja) suffered a severe injury to his left forearm when he was 3 months old; he was therefore excluded from the analysis. Four individuals for whom fewer than six bouts were obtained were also excluded from individual analyses. This left us with a total of 37 individuals.

Testing procedure

A PVC tube (length, 10 cm; diameter, 3 cm) with a favorite food (mashed sweet potato) smeared inside its edges was given to the subject or placed on the floor of the enclosure, and the hand (and finger) used to extract the food was recorded (Fig. 1). The hand used by the experimenter to offer the tube was recorded and shown to have no effect on hand preference [Pearson's $r = 0.04$, $t(95) = -0.45$, $p = 0.65$]. Subjects performed the task either sitting on a substrate, with both hands available for tube manipulation (37 individuals; mean number of bouts, 33.84, range 6–128), or hanging, i.e., supporting their body weight with one hand and using the other hand to extract food, while holding the tube with one or both feet (12 individuals; mean bouts, 15, range 6–40).

Table 2 Species and sex of subjects of current study

Species	F	M
S. syndactylus	10	12
Hylobates lar	4	4
Hylobates muelleri	2	2
Hylobates pileatus	2	1
Hylobates agilis × *Hylobates albibarbis*		2
Hylobates agilis	1	
Hylobates albibarbis		1
Nomascus leucogenys	1	
Total	20	22

F Female, *M* male

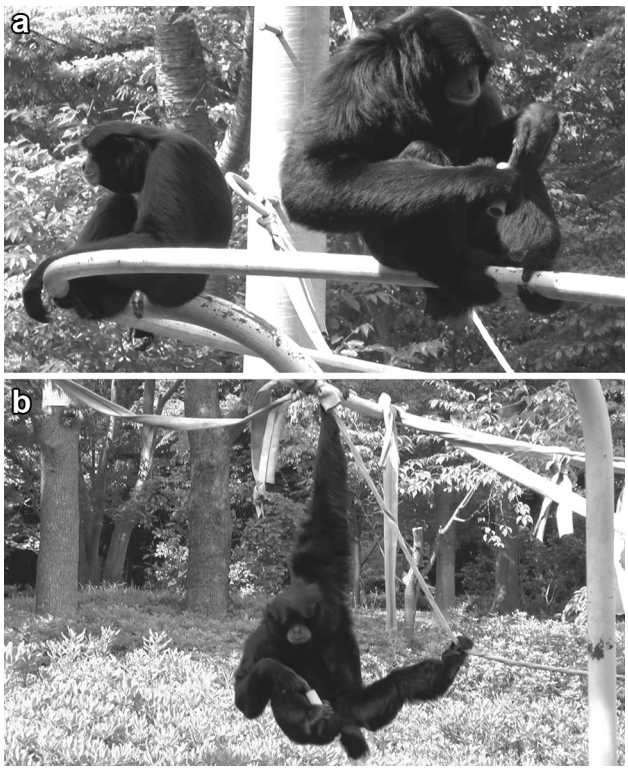


Fig. 1 Subjects performing the tube task while sitting (**a**) or hanging (**b**)

Data analysis

L. M. and M. U. recorded all experiments with video cameras (Panasonic HC-X920M or Sony HDR-CX560V). L. M. recorded hand use frequency (the raw number of times an individual inserted a finger into the tube and brought it to the mouth) and bout frequency from the videos. A new bout was recorded whenever the subject dropped the tube, changed its position of the hand, held the tube with both hands, or switched hands. Following standard procedures (e.g., Hopkins et al. 2007), we calculated a frequency handedness index (HI-Freq.) for each individual by subtracting the total number of left-hand uses from the total number of right-hand ones, and dividing by the total number of hand uses for that individual. In the same way, we calculated a bout handedness index (HI-Bout). When more than five data points were available, we determined individual hand preference on the basis of binomial *z*-scores (calculated from the frequency of use of left and

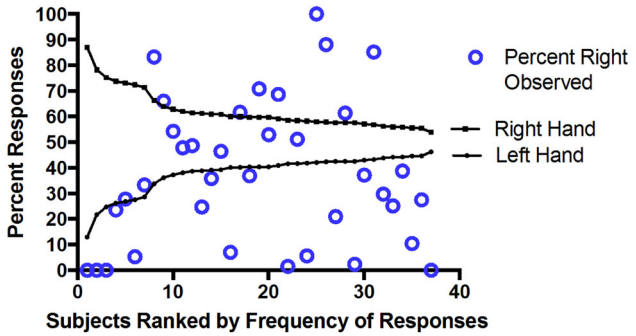


Fig. 2 Funnel plot for our dataset. If sampling bias occurs, a higher proportion of lateralized subjects would fall at the front of the funnel and fewer at the tail, which is not the case here

right hand). Following Marchant and McGrew (2007), we classified individuals with scores higher than 1.96 as left-handed, individuals with scores lower than −1.96 as right-handed, and individuals with scores between −1.96 and 1.96 as ambiguously handed.

We used Pearson's product-moment correlations to test for any relationship between sample size and hand preference and to compare results based on raw frequencies and bouts. We inspected the variation in the number of responses across subjects by means of a funnel plot (Palmer 2002): visual inspection of the funnel plot revealed no sampling bias, with many lateralized subjects falling at the tail end of the funnel plot and non-lateralized subjects evenly distributed in the funnel (Fig. 2). We used χ^2 and inferential statistics to determine if there was a significant hand preference and to compare the results on these captive apes to those of Morino (2011) on wild siamangs. All tests were two-tailed, with the significance level set at $p < 0.05$.

We constructed two sets of linear mixed-effects models (LMM) and used multi-model inference and model averaging to evaluate factors that could affect the expression of hand preference (HI). In the first set we tested species, sex, age and posture; in the second one we tested species, sex, age and the use of different fingers. We could not evaluate all variables at once, as we only had data on finger use for the sitting posture. We set individual identity as a random effect in all models to control for repeated measures. We ran the model selection using the R package MuMIn (Bartoń 2013; R version 3.3.2, R Core Team 2014). We ranked candidate models according to their Akaike information criterion (Akaike 1985) corrected for small sample sizes and their normalized Akaike weights (Burnham and Anderson 2002). We used the full model average method to calculate the relative weight of evidence for each predictor involved in the models, by summing the model weights over all models in which each term appeared (Symonds and Moussalli 2011). We calculated the 95% confidence interval for each predictor, and treated them as significant if their 95% confidence interval did not include zero.

Results

Hand preference in tube task

First, we tested for population-level biases using one-sample *t*-tests on the HI-Freq. and HI-Bout scores. Raw frequencies and bout numbers are given in Table 5 (Appendix). There was no significant correlation between sample size and HI (Pearson's $r = 0.16$, $t(35) = -0.99$, $p = 0.33$). Significant leftward asymmetries in hand use were found for both measures [Hi-Freq, mean ± SE = −0.25 ± 0.09, $t(36) = -2.73$, $p < 0.01$; Hi-Bout, mean ± SE = −0.18 ± 0.08, $t(36) = -2.22$, $p = 0.03$; Table 3]. The individual HI scores for each individual in the sitting posture are shown in Fig. 3. Since the HI-Freq. and HI-Bout data were significantly positively correlated (Pearson's $r = 0.94$, $t(35) = 15.97$, $p < 0.001$), we only considered hand use frequency data for the remaining analyses. Subjects were classified as 20 left-, eight as ambiguously and nine as right-handed (see Table 3 for classifications within different categories). Consistent with the HI-Freq. results, there were significantly more left- than right- ($n = 29$; $\chi^2 = 4.17$, $p = 0.04$) and ambiguously handed ($n = 28$; $\chi^2 = 5.14$, $p = 0.02$) apes in our overall sample.

Table 3 Hand preference frequency (based on z-scores calculated from frequency data) and mean frequency handedness index (*HI-Freq.*) scores with SE for different experimental conditions of the tube task

Data subset	No. L	No. R	No. A	Mean HI-Freq.	SE
Sitting condition	20	9	8	−0.25	0.09
Hanging condition	6	2	4	−0.18	0.16
Siamangs (sitting)	10	2	8	−0.33	0.11
Non-siamangs (sitting)	10	7	0	−0.16	0.16
Index finger (all, sitting)	16	5	8	−0.31	0.11
Thumb (all, sitting)	9	7	9	−0.20	0.07
Wild siamangs[a]	22	10	4	−0.27	0.12

L Left-handed, *A* ambiguously handed, *R* right-handed

* $p < 0.05$; ** $p < 0.01$

[a] data from Morino (2011)

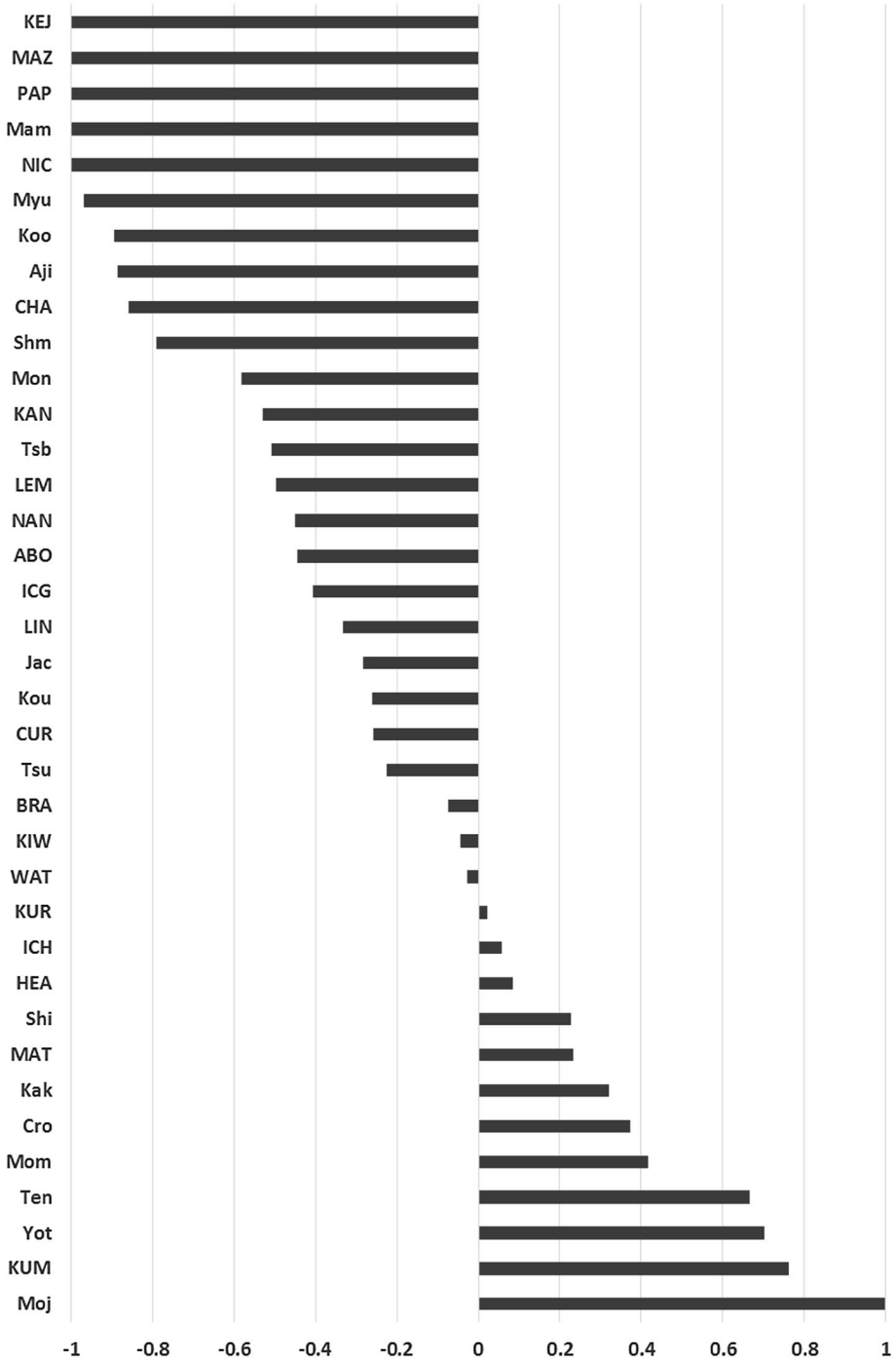


Fig. 3 Distribution of individual frequency handedness index (calculated from raw frequencies in the sitting condition). A value of -1 indicates exclusive use of the left hand, *zero* indicates complete absence of preference. Names of siamangs are *capitalized*

Based on our multi-model inferencing and model averaging, finger use significantly affected the expression of hand preference: although subjects used the index finger or the thumb in equal proportions (index 56% of the time), using the index finger revealed a significantly stronger left-hand preference than using the thumb (Table 4; Fig. 4). Species (siamang vs. other gibbons), sex, posture (sitting vs. hanging) and age (mature vs. immature) had no significant influence on hand preference (Table 4).

Comparison with wild siamangs

The average HI and proportion of left-handed siamangs in this study did not differ significantly from those of the wild

Table 4 Model-averaging results

Model/predictor	Estimate ± SE	2.5% CI	97.5% CI	W evidence
Model 1[a]				
Species	−0.232 ± 0.180	0.586	0.122	0.38
Sex	−0.078 ± 0.180	−0.432	0.275	0.17
Age	−0.181 ± 0.220	−0.611	0.250	0.22
Posture	−0.136 ± 0.144	−0.418	0.147	0.28
Model 2[b]				
Species	−0.203 ± 0.180	−0.557	0.149	0.37
Sex	0.059 ± 0.180	−0.296	0.413	0.16
Age	−0.073 ± 0.222	−0.508	0.362	0.16
Finger	*0.254 ± 0.120*	*0.020*	*0.489*	*0.81*

Model averaging using Akaike information criterion. Estimates, SE, 95% confidence interval (95% CI) and the relative weight of evidence (*W evidence*) for each predictor are presented. Predictors with 95% CI that do not overlap with 0 (significant influence on the response variable) are shown in *italic*

[a] HI ~ species + sex + age + posture + (1|individual ID)

[b] HI ~ species + sex + age + finger + (1|individual ID)

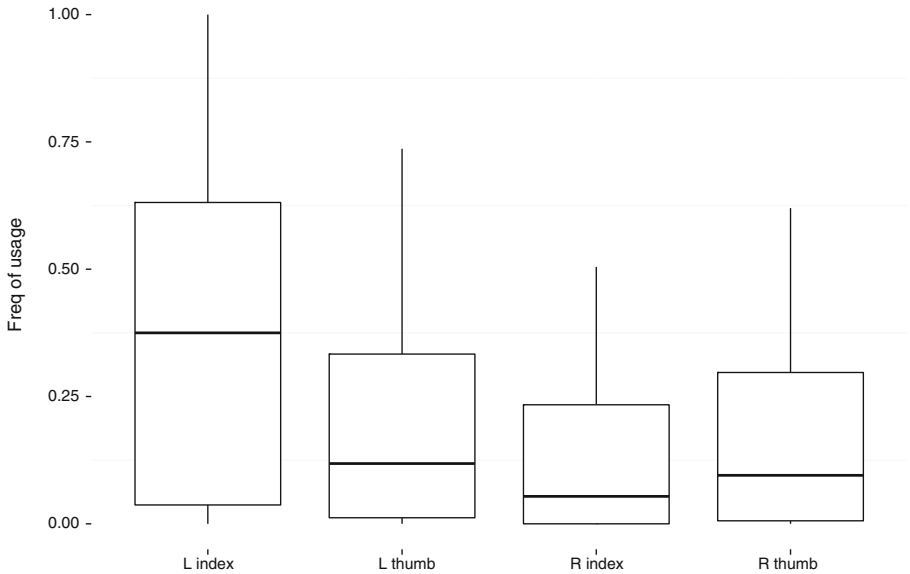


Fig. 4 Percentage of each hand/digit combination when performing the tube task. *L* Left, *R* Right

siamangs studied by Morino (2011) [HI, $t(62.94) = 0.46$, $p = 0.65$; left-handed proportion, $\chi^2 = 1.15$, $p = 0.28$; Table 3].

Discussion

Hand preference and the PO hypothesis

The hylobatids tested in the tube task showed a significant hand preference at the individual level (78% showed a significant preference), and a significant left-hand preference at the population level (69% of the lateralized individuals). This is in accordance with predictions of the PO hypothesis for such a highly arboreal taxon (MacNeilage et al. 1987), and is consistent with previous findings on wild siamangs (Morino 2011). While our LMM analysis ruled out any effect of species, caution is required when interpreting these results. The frequency of left/right individuals was 10/2 for siamangs, and 10/7 for gibbons (Table 2), so for gibbons separately, neither the frequency of left individuals nor their average HI differs from chance. The lack of a significant impact of the factor species on hand preference could be due to the associated high variation, which in turn might be because we contrasted the larger-bodied siamangs—accounting for approximately

half the sample—versus smaller gibbons belonging to six distinct species. The rationale for lumping all gibbon species was that they have roughly the same body size—much smaller than siamangs—and a comparable degree of arboreality and basic ecology, but it is possible that other factors make this grouping inappropriate. For these reasons, as well as the lack of consistency across earlier studies on captive gibbons (Olson et al. 1990; Stafford et al. 1990), we recommend further research on smaller gibbons to confirm the results of the present study and explicitly investigate variation within the hylobatid family.

Task complexity and digit use

Many authors have underlined the importance of task complexity in manual laterality (Fagot and Vauclair 1991; Blois-Heulin et al. 2006). Our finding of a significantly stronger hand preference when using the index finger than when using the thumb could be interpreted in this light: introducing the index finger deeper into the interior of the tube, which is less visible and harder to reach, might be motorically and cognitively more complex than using the thumb to remove food at the edge of the tube, and is thus more likely to elicit a hand preference. Recent studies reported different levels of preference when using different fingers for the tube task (Nelson and Boeving (2015) on spider monkeys; Maille et al. (2013) on guenons and mangabeys), and such a difference could also be expected in hylobatids given the position of their thumb.

Alternatively, there could be a shift toward the right hand associated with the use of the thumb. This has been documented for great apes, in which precision grips involving the thumb were shown to be primarily associated with right-hand use (Christel 1994; Tonooka and Matsuzawa 1995; Hopkins et al. 2002; Hopkins and Russell 2004; Pouydebat et al. 2011; Meguerditchian et al. 2015). Due to the position of their thumb, hylobatids have a more limited range of grips than other apes or cercopithecids (Christel 1994). However, the thumb is strongly involved in object manipulation, especially food items (Lorenz 1974; Prime and Ford 2016), and it is thus possible that its use could trigger a shift in hemispheric involvement.

Further research on hylobatids should investigate the implications of using the thumb vs. the index finger, and whether different grips and/or postures are associated with use of, or preference for, a specific hand. Another promising approach, exploring a different facet of laterality, would be to compare hand and grip preference in terms of efficiency, e.g., of speed of movement or number of errors (e.g., Hopkins et al. 2002; Spinozzi et al. 2004), or strength of grip.

Wild-captive comparison

Compared to their wild counterparts, the captive siamangs of this study differed in two major ways: a much higher proportion of ambiguously handed individuals (40 vs. 11% in the wild); no age-related variation, as opposed to significantly higher preference scores in wild adults compared to immatures (Morino 2011). The main difference in the postural life of hylobatids in captivity compared to the wild is the time they spend on the ground (Morino, personal observation). On the one hand, this could relax the selective pressure on a strong hand to support body weight, and on the other, increase opportunities to use both hands, which are more often free. These two, not mutually exclusive, factors could explain the higher proportion of ambiguously handed individuals, and the smaller difference between adults and immatures in captive conditions. Nevertheless, in our sample there was no difference in hand preference between the hanging and sitting conditions. More data on young individuals and other species of different size and degree of arboreality could help address these conflicting results. A related research topic would be to examine why some individuals retain arboreal habits in captivity, while others seem content to remain on the ground.

Conclusion

The captive siamangs in this study displayed a significant left-hand preference, confirming previous findings on a wild population, and lending support to the PO hypothesis. The higher incidence of ambidextrous individuals among captive compared to wild animals suggests that factors such as fatigue and choice of supporting arm could be important for the expression of handedness in hylobatids. In accordance with current theory, a more complex motor action (i.e., the use of the index finger instead of the thumb) elicited a more lateralized response. More data on smaller hylobatid species are needed to confirm the pervasiveness of the left-hand bias across this family. Data on other highly arboreal primates—such as spider monkeys—could allow valuable comparisons, and a means of understanding the factors that initially selected for lateralized behaviors in the order Primates.

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and support of Mami Shikuwa at PRI, and the collaboration, enthusiasm and dedication of the keepers and administrators of the Japanese zoos and facilities where the data were collected (Table 5, Appendix). We thank Tatsuya Kurokawa, Hikaru Wakamori and Ryuji Yasukochi for their excellent data collection, and Peter MacNeilage, Adrien Meguerdichian, Shelly Masi and the participants of the psychology and the ecology seminars at PRI for helpful and stimulating discussions.

Compliance with ethical standards

The design of the present research was approved by the Animal Welfare and Animal Care Committee of the Primate Research Center (Kyoto University) and by the Animal Research Committee of Kyoto University (approval nos. 2013-173 and 2014-011). All procedures adhered to the Japanese Act on the Welfare and Management of Animals.

Conflict of interest The authors declare that they have no conflict of interest.

Appendix

See Table 5 in appendix.

Table 5 Attributes of study subjects, frequency (*Freq*.) and number of bouts (*Bout*) for the sitting condition

ID	Species	Sex	Age	Freq.	Bout	Location
Abo	Ss	M	A	18	6	J
Ajiru	Ha	F	A	142	18	J
Brave	Ss	M	A	92	39	C
Chan	Ss	M	O	119	26	I
Cromartie	Hl	F	A	115	15	K
Curly	Ss	M	A	224	48	S
Heart	Ss	F	A	66	11	C
Ichi	Ss	F	J	134	53	O
Ichigo	Ss	F	A	249	65	J
Jack	Hl	M	A	98	19	J
Jean[a]	Ss	M	A	1	1	K
Kan[a]	Hp	M	A	2	2	J
Kanako	Hp	F	A	50	17	J
Kanna	Ss	F	A	17	11	K
Keiji	Ss	M	A	52	6	N
Kiwi	Ss	M	SA	67	19	J
Koori	Hp	F	J	30	10	J
Kouki	Hm	M	SA	104	36	N
Kuro	Ss	F	A	255	75	S
Kuromi	Ss	F	A	182	67	I
Lemon	Ss	M	A	299	56	O
Lily[a]	Hl	F	O	4	1	K
Linda	Ss	F	A	21	10	J
Mamy	Hb	M	O	663	68	P
Matsu	Ss	F	A	238	62	I
Matsu Nag	Ss	F	A	105	19	N
Momo	Hl	F	J	108	46	S
Monjirou	Nl	F	O	153	31	G
Monrey	Hl	F	A	182	48	S
Myu	Hm	F	O	133	9	N
Nana	Ss	F	A	331	128	O
Nichi	Ss	M	I	23	7	O
Papaya[a]	Ss	M	J	7	3	J
Peach	Ss	F	J	169	43	J
Raja[a]	Hx	M	A	464	107	P
Shiro	Hm	M	A	168	29	N
Shiromartie	Hl	M	A	324	96	K
Tena	Hl	M	A	36	9	G
Tsubasa	Hm	F	A	96	49	N
Tsuyoshi	Hx	M	A	287	55	P
Wataru	Ss	M	J	112	43	S
Yotaro	Hl	M	A	209	75	S

ID Identity, *Ha Hylobates agilis*, *Hb Hylobates albibarbis*, *Hl Hylobates lar*, *Hm Hylobates muelleri, Hp Hylobates pileatus, Hx Hylobates agilis* × *Hylobates albibarbis*, *Nl Nomascus leucogenys*, *Ss Symphalangus syndactylus, C* Chiba Zoological Park (Chiba), *G* Hirakawa Zoological Park (Kagoshima), *H* Higashiyama Zoo and Botanical Gardens (Nagoya), *I* Itozu-no-mori Zoological Park (Kitakyushu), *J* Japan Monkey Centre (Inuyama), *K* Kyoto City Zoo (Kyoto), *O* Tennoji Zoo (Osaka), *P* Primate Research Institute (Inuyama), *S* Kujukushima Zoological and Botanical Gardens (Sasebo), M male, F female, *A* adult, J juvenile, SA subadult, *O* old, *Freq*. total number of hand uses

[a] Excluded from the analysis (see text for details)

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